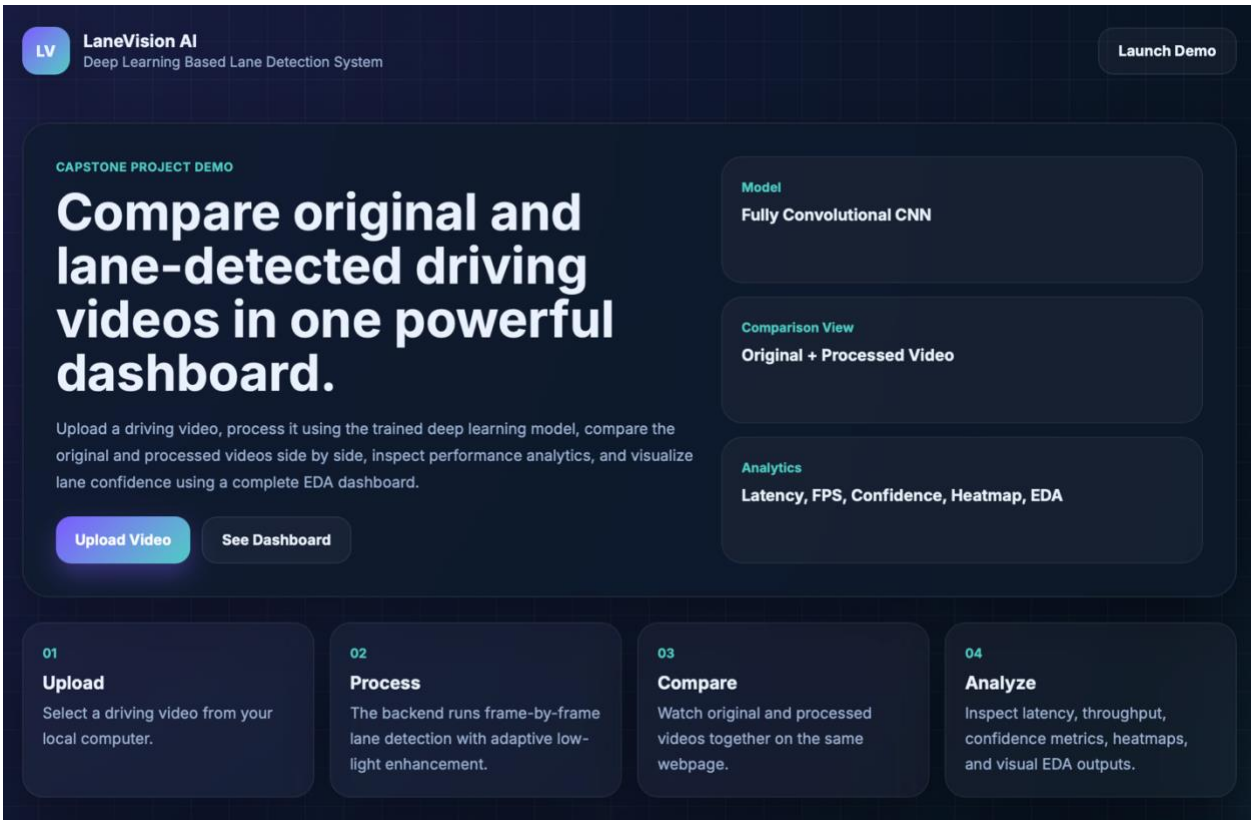
Project Title: Lane Detection Using Deep Learning
Course: DATS 6501 – Data Science Capstone

Summary

This project focuses on building an advanced lane detection system for driving videos using a deep learning approach. Since Progress Report 2, the project has evolved from a basic web-based application into a more advanced, interactive, and analytics-driven system.

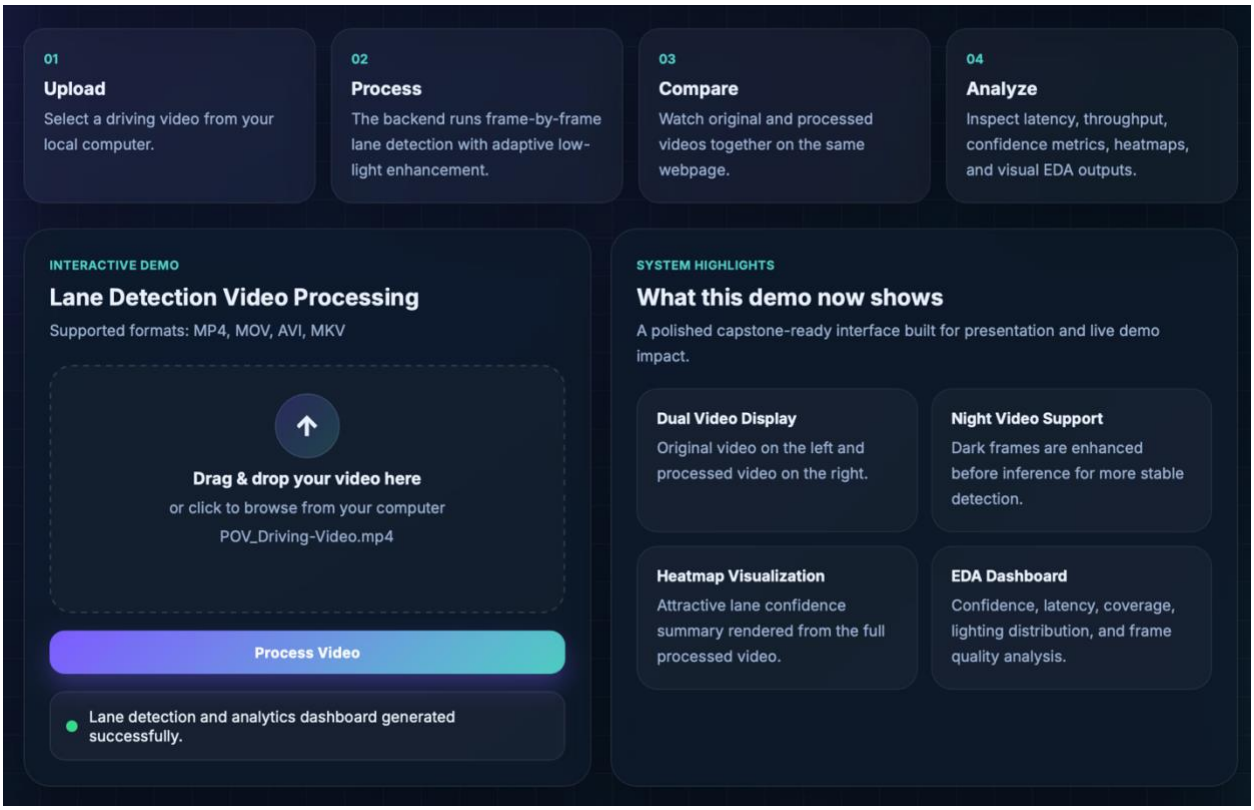
The focus of this phase was to enhance the system by introducing real-time style processing, dual video comparison, and a comprehensive analytics dashboard. The application now not only processes videos but also provides detailed performance insights such as latency, frame rate, lane confidence, and heatmap visualizations. These improvements significantly enhance both the usability and presentation quality of the project, making it closer to a production-level demonstration.



B. Analytics Integration

• Developed a full analytics dashboard to display processing insights.
• Added metrics such as:

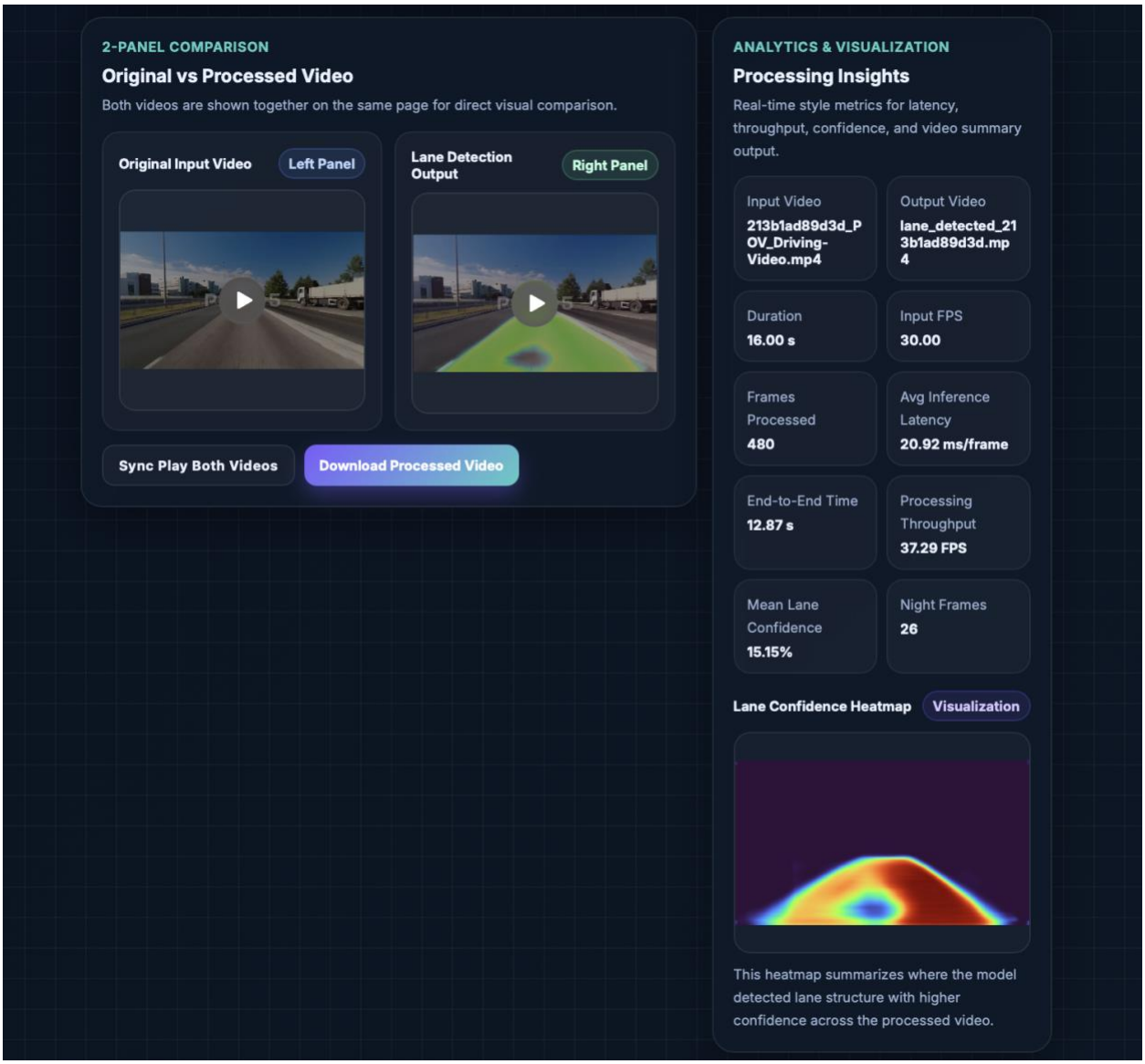
- Frame-wise inference latency
- Lane coverage distribution
  • Implemented automatic generation of visualization charts:
- Confidence trend graph
- Latency graph
- Histogram for lane coverage
  • Added summary statistics for both day and night frames.



C. Dual Video Comparison System

• Implemented a **2-panel layout** for simultaneous viewing of:

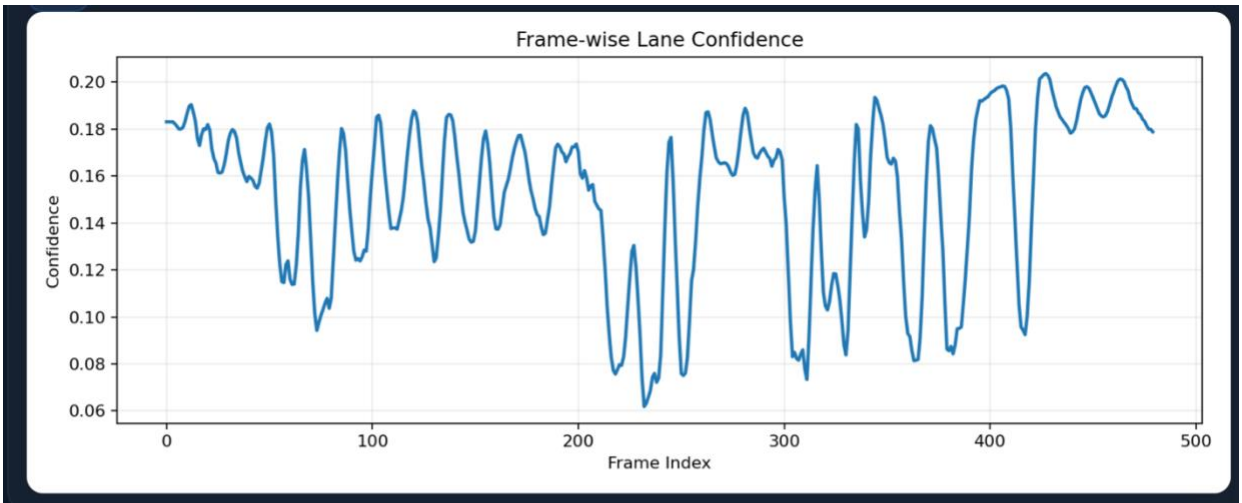
- Original input video
- Lane-detected output video
  • Enabled synchronized playback for better comparison.
  • Improved visual clarity of lane overlays using heatmap blending.
  • Enhanced user experience by placing both videos in a single dashboard view.



D. EDA (Exploratory Data Analysis) Dashboard

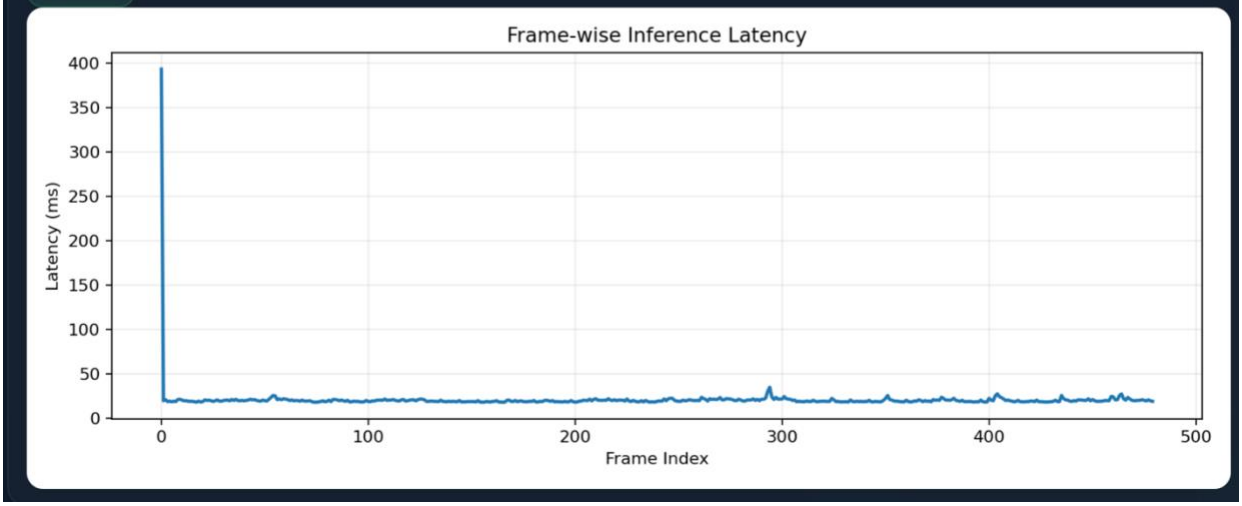
• Built a detailed EDA panel to explain model behavior.
• Included:

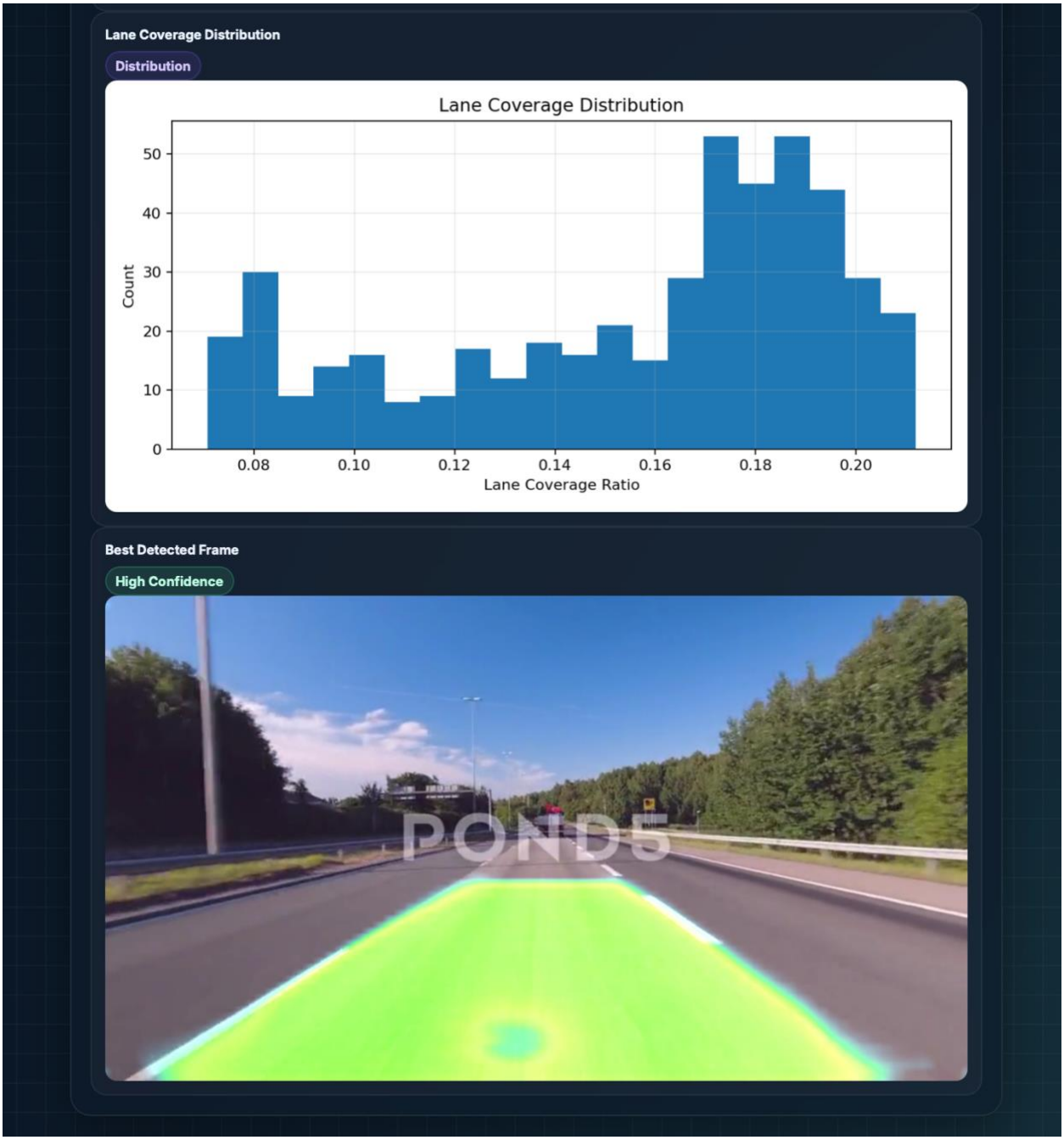
- Frame-wise confidence plot
- Frame-wise latency plot
- Lane coverage distribution histogram
- Best detected frame visualization
  • Provided a structured explanation of model performance across the entire video.



Frame-wise Latency

Performance





Challenges Encountered

• Maintaining real-time performance while processing video frame-by-frame was challenging.
• Integrating analytics generation without slowing down processing required optimization.
• Managing multiple outputs (video, charts, heatmaps) increased system complexity.

Next Development Milestones

• Add advanced evaluation metrics such as IoU.
• Enhance UI with more interactive controls for analytics exploration.
• Prepare final presentation and documentation for submission.

Conclusion

In summary, the project has progressed from a basic web-based lane detection system to a fully interactive analytics-driven application. The addition of dual video comparison, real-time performance tracking, and EDA visualization significantly enhances the system's functionality and presentation.

The project now demonstrates not only the ability to detect lanes but also to analyze and explain the model's behavior, making it a strong and complete capstone submission.